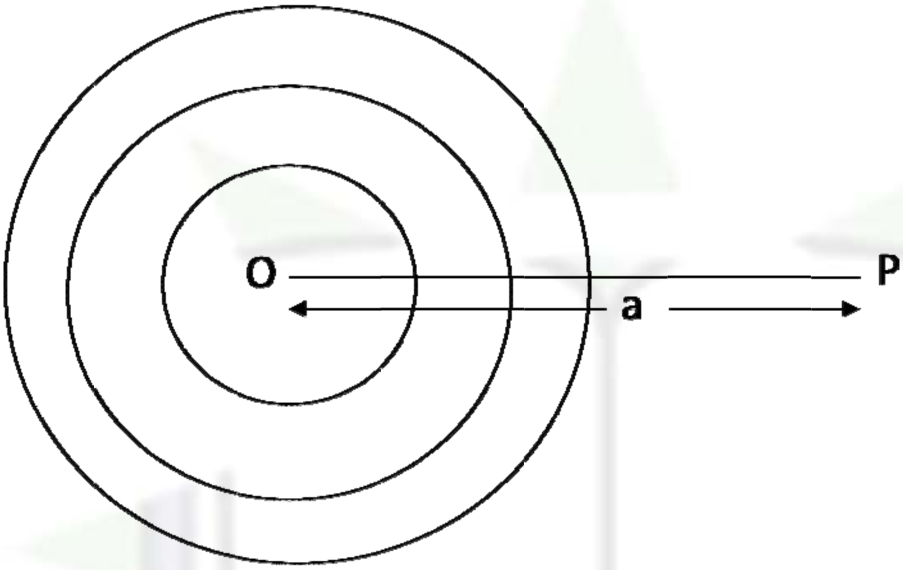
Gravitational Potential at a point outside a solid sphere:

Given M & R = Mass & Radius of the solid sphere

OP = a = the distance of the given point from the center of the solid sphere a>R

We can imagine the solid sphere to be made up of large no of hollow thin concentric spherical shells of masses m_1, m2, m3…



Since the given point P lies outside these shells hence potential at P due to these thin shells

$$-\frac{Gm_1}{a}, -\frac{Gm_2}{a}, -\frac{Gm_3}{a}, ...$$

Hence total potential at P due to these thin shells i.e. due to the solid sphere

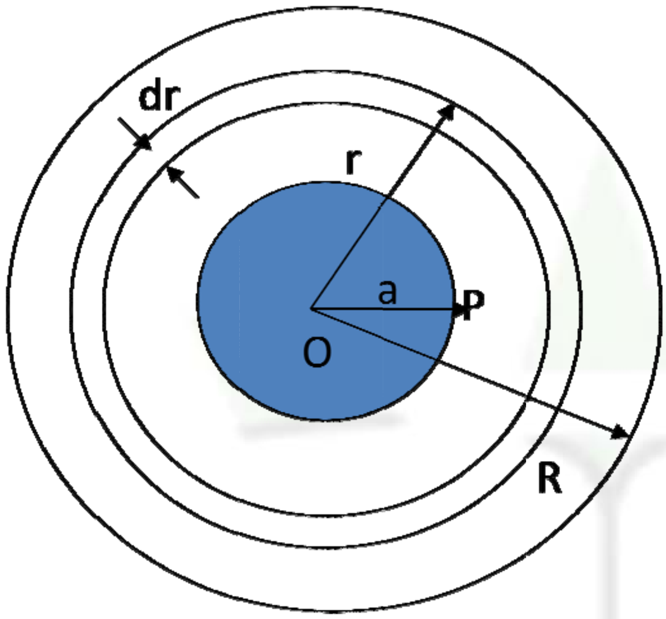
$$V = -\frac{Gm_1}{a} - \frac{Gm_2}{a} - \frac{Gm_3}{a} + ... = -\frac{G}{a}\left[m_1 + m_2 + m_3 + ...\right] = -\frac{GM}{a}$$

If we assume a point mass M equal to the mass of the solid sphere at the center O of the sphere potential at P

$$V = -\frac{GM}{a}$$

Thus to find the potential at any external point we can imagine the mass of the solid sphere to be concentrated at its center.

Gravitational Potential at a point inside a solid sphere



Given M & R = Mass and Radius of the solid sphere

OP = a = The distance of the given point from the center of the solid sphere (a < R)

Mass per unit volume $= \dfrac{3M}{4\pi R^3}$

Let us imagine that the solid sphere is made up of two parts

(1) The inner solid sphere with center at O & of radius 'a' passing through the given point P.

(2) The concentric outer thick shell having thickness (R – a)

Let V_1 & V_2 be the potential at the point P due to the inner solid sphere and the outer thick shell respectively.

Total potential at P $V = V_1 + V_2$ ⟶ (1)

<u>To find V_1</u>

The volume of the inner solid sphere $= \dfrac{4}{3}\pi a^3$

Mass of the inner solid sphere $= \dfrac{3M}{4\pi R^3} \times \dfrac{4}{3}\pi a^3 = \dfrac{Ma^3}{R^3}$

Since the given point P lies on the surface of the inner solid sphere the potential at P due to it

$$V_1 = -\frac{G \times mass}{Radius} = -\frac{G \times \frac{Ma^3}{R^3}}{a} = -\frac{GMa^2}{R^3} \longrightarrow \quad (2)$$

<u>To find V_2</u>

We can imagine the outer thick shell to be made up of large no of concentric thin shells.

Let us consider one such thin shell of radius $r(a \le r \le R)$and radial thickness dr

The volume of the material of the thin shell = Surface area of the shell X thickness

= $4\pi R^2 dr$

Mass of the elementary shell $= \frac{3M}{4\pi R^3} \times 4\pi r^2 dr = \frac{3M}{R^3} r^2 dr$

Since the given point P lies inside the thin shell the potential at P due to the thin shell will be same as the potential on the surface of the shell.

Potential at P due to the elementary thin shell

$$dv_2 = -\frac{G \times (mass)}{Radius}$$

$$dv_2 = -\frac{G \times 3Mr^2dr/R^3}{r} = -\frac{G3Mrdr}{R^3} \longrightarrow \quad (3)$$

Hence the total potential at P due to the thick shell can be obtained by integrating (3) between the limits r=a to r=R

$$V_2 = \int dv_2 = \int_a^R \frac{-G3M}{R^3} rdr = \frac{-G3M}{R^3}\left[\frac{r^2}{2}\right]_a^R$$

$$V_2 = -\frac{GMa^2}{2R^3}\left[R^2 - a^2\right] \longrightarrow \quad (4)$$

Putting equation (2) and (4) in (1)

$$V = -\frac{GMa^2}{R^3} - \frac{G3M}{2R^3}(R^2 - a^2)$$

$$V = -\frac{GM}{2R^3}(3R^2 - a^2) \longrightarrow \quad (5)$$